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Lab 3: Component Modelling & Architectural Pattern Selection

Architecture Selection - Personal Subscription & Recurring Expense Auditor

Architecture Selection

We chose Layered Architecture for the Personal Subscription & Recurring Expense Auditor.

Architectural Choice

The system is divided into Presentation, Business, and Data layers. The Presentation Layer contains the User Interface Component. The Business Layer contains the Subscription Manager Component, Recurring Expense Auditor Component, and Payment & Transaction Service Component. The Data Layer contains the Database Component.

Reason 1

Layered architecture provides a clear separation of responsibilities. User interaction, subscription management, recurring-expense auditing, transaction processing, and data storage are handled by separate components. This makes the system easier to understand, maintain, and modify.

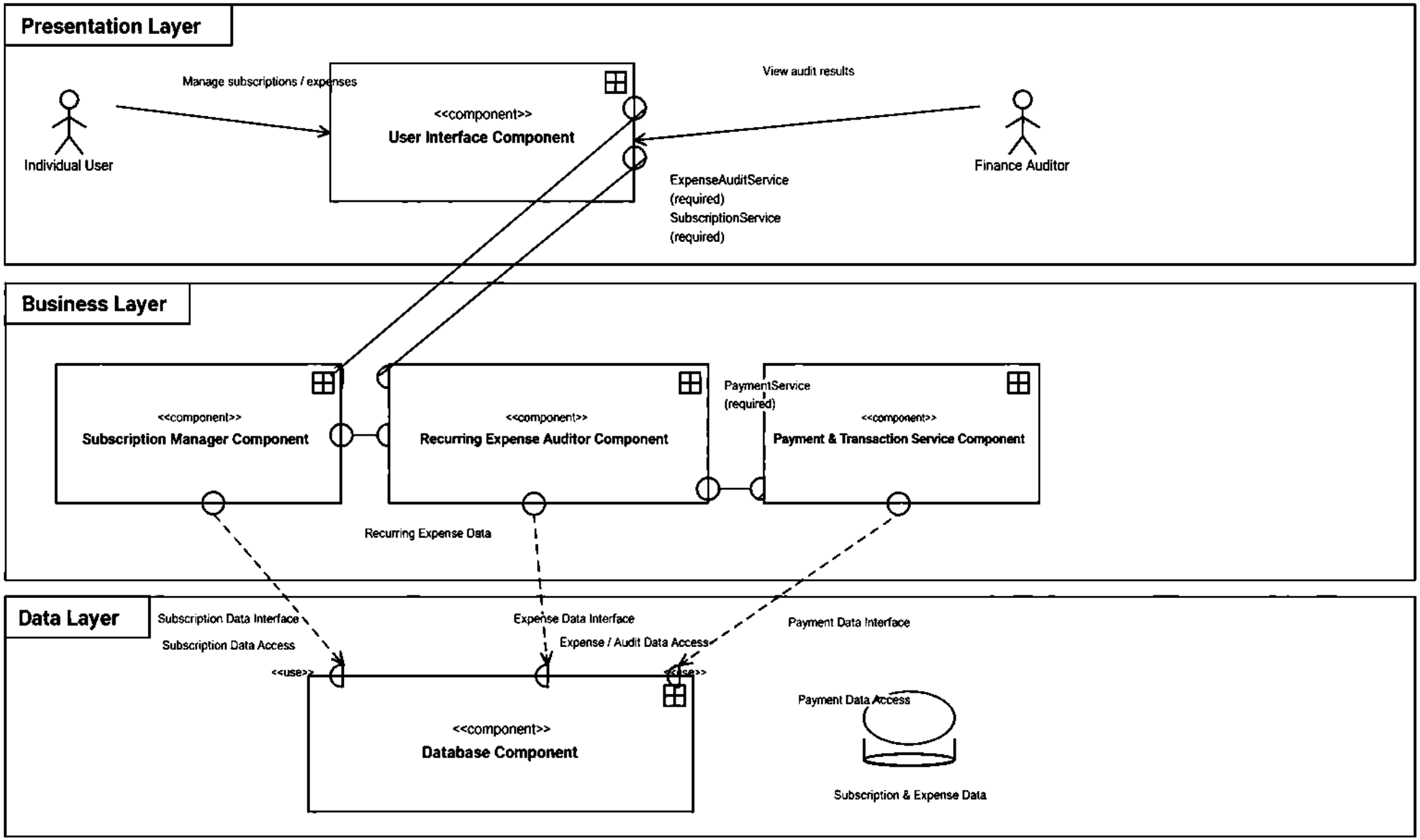
Reason 2

The architecture is suitable for the Personal Subscription & Recurring Expense Auditor because the system has several related financial functions that can communicate through well-defined interfaces. It provides a simple and organized structure without the additional operational complexity of a microservices architecture.

Security Advantage

The layered design allows sensitive financial information and transaction processing to be handled through business-layer components rather than allowing the User Interface to directly access the database. This provides better control over access to sensitive resources.

Performance Benefit

The separation of responsibilities allows frequently requested operations, such as checking subscriptions and identifying recurring expenses, to be handled by the Recurring Expense Auditor and supported by the database without placing unnecessary processing on the User Interface. This helps maintain efficient request processing.

Layered Architecture